



MONASH University

Assessment

Faculty of Information Technology

FIT5125

IT Research and Innovation Methods

Semester 1 2026

Assignment 1

Value

This assignment is worth **45%** of the total marks for FIT5125

Assignment due date

Thursday, 23 April 2026, 11:55 PM (Melbourne time)

Submission method

Submit to Moodle:

- Multiple files as specified in the description of each of the three tasks.

Assignment Criteria

Task A: Research Ethics	Task B: Questionnaires	Task C: Interviews
Weighting: 15 marks	Weighting: 15 marks	Weighting: 15 marks
See Page 3.	See Page 4.	See Page 5.

This is an individual assessment; it must be your own work and expressed in your own words.

 *A marking guide is available on Moodle.*

 *There are specific requirements for file names on your submission (see the instructions for each task).*

Assessment rules:

1. Note that plagiarism detection procedures may be applied to each submission. See the University rules and regulations regarding plagiarism and resulting penalties. Any case of plagiarism detected will mean automatic failure of the entire assignment. Note that even where Turnitin reports a high similarity score, this may simply be the result of text that is part of the original question or a correctly referenced quotation (such cases are not plagiarism).
2. Late submissions will incur a penalty of 5% per day, see:
<https://publicpolicydms.monash.edu/Monash/documents/1935752>
3. Submissions should follow the instructions for each task, described under "What to Submit".
4. Monash policy on Special Consideration is available at:
<https://www.monash.edu/exams/changes/special-consideration>
5. We aim to mark and return work within 10 working days of the submission due date.

Task A

“Planning for ethical research”

Consider the following scenario:

You work for the product team of an Australia-based company building a SmartC mobile app that integrates with university systems. The app uses AI to predict student wellbeing risks (e.g., burnout and disengagement) and provides early interventions such as nudges, support resources, or alerts.

The system combines multiple data sources, including academic activity (e.g., LMS usage and class attendance), location data (on-campus movement patterns via Wi-Fi), wearable device data (e.g., sleep and activity levels, if users opt in), and communication patterns (e.g., app messages and help requests).

The AI system generates risk scores and may recommend interventions to students. In some cases, the AI system may notify university support services.

Your company is preparing to conduct a study to evaluate the app's effectiveness in identifying at-risk students and improving wellbeing outcomes. The study will use:

- Aggregated behavioral and location data
- AI-generated risk scores and intervention logs
- Optional wearable device data
- Interviews with students about perceived accuracy, fairness, and intrusiveness

The study will follow the 4-stage research pipeline:

1. Recruitment
2. Data Collection
3. Data Analysis
4. Dissemination

If successful, the findings will be used to promote the app to universities as a “predictive student wellbeing solution.”

Your Task

Your task is to assess potential ethical risks in this study. You are required to create a four-slide presentation that shows:

- The two most significant ethical issues at each stage of the research pipeline.
- The mitigation strategies you would recommend to address each of these ethical risks.

Prepare your responses

- Identify Ethical Issues:

For each stage (Recruitment, Data Collection, Data Analysis, Dissemination), choose two key ethical concerns that are relevant to a corresponding stage of the study.

Clearly describe each concern.

- Propose Mitigation Strategies:

For each identified ethical concern, provide a clear and relevant strategy that your team would implement to eliminate the risk.

Create a Four-Slide Presentation

Slide 1: Recruitment – Two ethical issues + mitigation strategies

Slide 2: Data Collection – Two ethical issues + mitigation strategies

Slide 3: Data Analysis – Two ethical issues + mitigation strategies

Slide 4: Dissemination – Two ethical issues + mitigation strategies

Ensure your slides are clear, concise, and well-structured (use bullet points where needed).

Submit Your Work

Prepare a PDF file with your four-slide presentation.

Upload your final presentation by the deadline on Moodle.

What to Submit

1. A PDF document, named "STUDENT-ID-Assignment-1A.pdf", containing your slides.

How Much to Write

Following your company's communication policy, each slide should have a maximum of 200 words (including titles).

What to Know

- If you have any questions about the assignment, you should submit them as a public post to the Ed forum (under the sub-category "Assignments") so that all students have access to the question and CE or admin tutor's response.
- This is an individual exercise that forms part of the formal assessment for the unit, you must therefore work alone and follow Monash University's policies, procedures and regulations relating academic integrity, plagiarism and collusion (see Moodle).
- As this is a formal assessment tutors are not allowed support to directly, however, they can provide feedback on any studio activities (during the studio).
- This task supports unit learning outcome 3 (Explain the ethical and professional issues that may arise when conducting IT research and innovation and demonstrate practical ways to address them).

Task B

"Designing a student survey to evaluate student experiences of the CoAlmate app"

Being able to design a survey to assess the beliefs, attitudes and experiences of users of technology is a key research skill that all IT professionals should have an understanding of. Consider the following scenario:

You work for the Teaching and Learning team at Monash University. Your team has been asked to evaluate whether the University should officially recommend CoAlmate, an AI-powered collaboration tool designed to support group assignments.

CoAlmate includes the following features:

- Task allocation suggestions based on students' prior activity and stated preferences
- Real-time summaries of group discussions
- Contribution tracking (e.g., number of messages, edits, and time spent in the platform)
- AI-generated participation scores for each student
- Automated suggestions to improve teamwork (e.g., prompting less active members, redistributing tasks)

As part of a structured trial, CoAlmate has been introduced to all students in one FIT unit for use in a graded group assignment.

All students were instructed to:

- Use CoAlmate for at least one full group assignment cycle
- Review their AI-generated participation score at least once
- Use at least one AI-generated suggestion during the project

System logs confirm that all students interacted with the platform at least minimally, but the level of engagement varies (low, medium, high usage).

Your task is to design a questionnaire to assess students' experiences and views on the app, focusing on:

- Usability
- Effectiveness
- Engagement
- Impact on Learning
- Likelihood of Continued Use

Your questionnaire must meet the following requirements:

- Use only closed-ended questions
- Include no more than 20 questions
- Make the questionnaire suitable for completion in no more than 10 minutes
- Make the questionnaire understandable without additional explanation

- Collect data that will help researchers determine student engagement levels (low, medium, high), and thus complement student engagement information obtained via log data

Since the survey is only for internal decision-making, you do not need to seek ethics approval from Monash University's Human Research Ethics Committee (HREC).

Your task

1. Create Your Questionnaire in Google Forms

- Use only closed questions
- Be concise (no more than 20 questions).
- Focus on key aspects of students' experience that are relevant to decision-making (usability, effectiveness, engagement, impact on learning, likelihood of continued use).

2. Write a rationale (max. 500 words) explaining

- Why you selected specific questions?
- Why you used certain closed question types and certain paths (if any)?
- How the survey structure ensures reliability and efficiency?

What to Submit

1. A PDF document, named "STUDENT-ID-Assignment-1B-survey.pdf", containing a "print" of your questionnaire (using the Google Forms "print" option).
2. A PDF document, named "STUDENT-ID-Assignment-1B-rationale.pdf", a written rationale of no more than 500 words justifying the design of your survey.

How Much to Write

Do not use more than 500 words for the written description of the rationale for the design of your questionnaire.

What to Know

- If you have any questions about the assignment, you should submit them as a public post to the Ed forum (under the sub-category "Assignments") so that all students have access to the question and CE or admin tutor's response.
- This is an individual exercise that forms part of the formal assessment for the unit, you must therefore work alone and follow Monash University's policies, procedures and regulations relating academic integrity, plagiarism and collusion (see Moodle).
- As this is a formal assessment tutors are not allowed support to directly, however, they can provide feedback on any studio activities (during the studio).
- This task supports unit learning outcome 1 (Demonstrate the ability to frame and design all aspects of the research and innovation pipeline) and unit learning outcome 4 (Evaluate, select and justify research and innovation methods, including data collection and analysis using qualitative and quantitative methods)

Task C

"Planning and conducting a research interview"

This assessment task aims to evaluate your knowledge and skills related to planning and conducting research interviews. Knowing how to conduct a research interview is not only valuable for conducting qualitative research, but for the many interviews in other settings that you will be required to conduct in your career. The ability to plan and conduct a high-quality interview is an important skill that all IT professionals require.

Consider the following extension to the scenario described in Task A:

Following the initial development of the SmartC app described in Task A, Monash University is now considering how the system should operate once students are identified as being at elevated wellbeing risk.

The app currently uses AI models to generate risk scores based on multiple data sources (e.g., academic activity, location patterns, wearable data, and communication signals). When certain thresholds are reached, the system can take different forms of action within the app or beyond it.

At this stage, the University has not yet finalised how the system should behave in these situations. There are multiple possible ways the system could respond, and these may affect students in different ways depending on the context.

The Monash University Student Advisory Council (MUSAC) has been asked to gather student perspectives to inform future decisions about how the system should operate in practice.

You have been contracted to conduct interviews with Monash students to explore their attitudes, expectations, and concerns regarding how such a system should function when it identifies potential wellbeing risks. Students may have different levels of familiarity with the app and may not have considered these issues before.

Your task

1. Create an interview schedule for a semi-structured interview with students attitudes, expectations and concerns about issues that might inform future decisions about how the system should operate in practice. Include an introduction in the interview schedule to explain the purpose of the interview, what's expected from the interviewee, and the interviewee's right as a research participant.
2. Create a rationale (no more than 500 words) for the design of your interview schedule explaining alignment of the interview schedule with the study objectives; and justifying the choice, arrangement, and types of questions
3. Recruit and interview a fellow student in English (from any unit, program or course at Monash University). Make sure that you create an appropriate consent form and that your interviewee signs the consent form before you interview them (you should not submit the signed form but keep it safe). Then, make sure that you record the interview.

 Your interview must be at least 15 minutes long.

What to Submit

1. A PDF document containing your interview schedule named "STUDENTID-Assignment-1C-schedule.pdf"
2. A description, of no more than 500 words, of your rationale for the design of your interview schedule named "STUDENTID-Assignment-1C-rationale.pdf".
3. An MP3 or MP4 audio file of the recording of your interview named "STUDENTID-Assignment-1C-recording.mp3"
4. A PDF document of your consent form (unsigned and not identifying your interviewee) "STUDENTID-Assignment-1C-consent-form.pdf"

How Much to Write

There is **no word limit for the interview schedule or for the consent form itself.**

There is a strict **500-word limit for the description of your rationale for the design of the interview schedule.**

What to Know

- If you have any questions about the assignment, you should submit them as a public post to the Ed forum (under the sub-category "Assignments") so that all students have access to the question and CE or admin tutor's response.
- This is an individual exercise that forms part of the formal assessment for the unit, you must therefore work alone and follow Monash University's policies, procedures and regulations relating academic integrity, plagiarism and collusion (see Moodle).
- As this is a formal assessment tutors are not allowed support to directly, however, they can provide feedback on any studio activities (during the studio).
- This task supports unit learning outcome 1 (Demonstrate the ability to frame and design all aspects of the research and innovation pipeline), unit learning outcome 2 (Review and critique the current state of art from a wide range of sources to inform a research question and appropriate methods), unit learning outcome 3 (Explain the ethical and professional issues that may arise when conducting IT research and innovation and demonstrate practical ways to address them) and unit learning outcome 4 (Evaluate, select and justify research and innovation methods, including data collection and analysis using qualitative and quantitative methods).